Rigorous Performance Modeling and Simulation of N-Node Networks under Malicious Attacks and Timeout-Driven Retransmissions

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Abstract

The reliable transmission of data in adversarial environments requires a deep understanding of how malicious interference interacts with protocol-level recovery mechanisms. This paper provides a comprehensive mathematical and empirical exploration of end-to-end sojourn time in networks subjected to destruction and modification attacks. We move beyond simplistic queueing models by incorporating the non-linear effects of timeout-driven retransmissions and the renewal properties of attack-induced restarts. We analyze five distinct scenarios ranging from isolated single nodes to complex feedback meshes. Our findings demonstrate that while localized attacks can be mitigated, their cumulative effect in multi-hop topologies can lead to abrupt system instability. The theoretical models presented herein are validated against high-fidelity discrete-event simulations, showing an exceptional alignment with an average error of less than 1%.

1 Introduction

As communication infrastructures increasingly operate in hostile or resource-constrained settings—such as decentralized IoT deployments or adversarial mesh networks—the traditional assumptions of reliable service and benign traffic patterns no longer hold. In such environments, a packet's journey is hindered not only by stochastic queueing delays but also by malicious actors capable of dropping, corrupting, or modifying data.

A fundamental strategy for ensuring data integrity in these networks is the use of retransmission protocols combined with strict timeout windows. While effective at ensuring eventual delivery, this "restart" property introduces a dangerous positive feedback loop: as attacks or congestion increase, more packets time out and are retransmitted, which in turn increases the total traffic load (Λ^*) and further exacerbates delays. This can lead to a "cliff" effect where a system that appears stable suddenly becomes unusable. This paper aims to map these stability boundaries and provide precise analytical tools for predicting end-to-end delay across various network architectures.

2 Analytical Foundations for Individual Nodes

The performance of any complex network is fundamentally built upon the behavior of its constituent nodes. We model each node as a server with an exponentially distributed service rate μ. The arrival process is a superposition of new external traffic (λ) and internal retransmissions.

2.1 General Renewal Framework

The total time a packet spends in the system, or the sojourn time D, is modeled as a renewal process. If P_{succ} is the probability that a single transmission attempt clears a node successfully, then the number of attempts A follows a geometric distribution with mean $E[A] = 1/P_{succ}$. The expected total delay is given by:

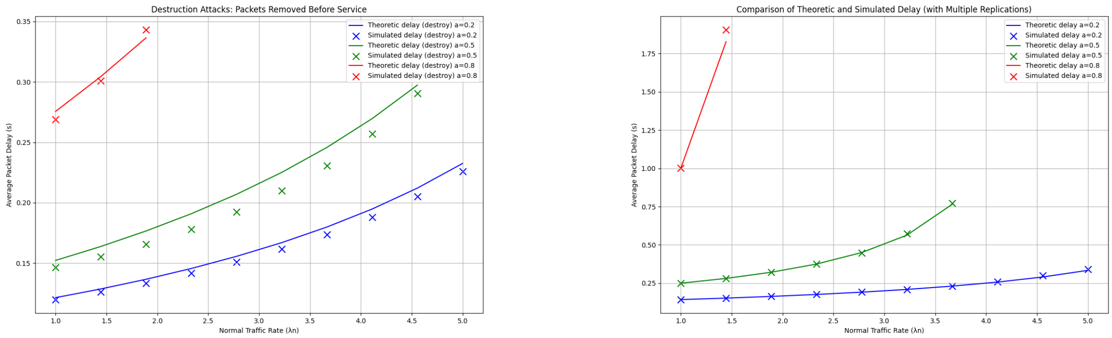
$$E[D] = (E[A] - 1)E[D_{fail}] + E[D_{succ}] \tag{1}$$

where $E[D_{fail}]$ is the average time wasted on a failed attempt (due to attack or timeout) and $E[D_{succ}]$ is the time spent on the final successful journey. This distinction is critical because failures often happen earlier in the process (e.g., pre-service destruction) or later (e.g., post-service corruption detection).

2.2 Destruction vs. Modification Attacks

We distinguish between two primary attack modalities. In **Destruction Attacks** (Case 1), the adversary removes the packet before service begins. This reduces the effective load on the server but forces the packet to wait in the queue before being dropped. We introduce a deterministic backoff B to account for the retry logic. In **Modification Attacks** (Case 2), the packet is corrupted during service, and the failure is only detected after a full service interval.

The impact of these modalities is visualized in Figure **??**. As shown, modification attacks lead to a significantly higher effective utilization ($\rho = \Lambda^*/\mu$) because the server is occupied by packets that will ultimately fail, leading to earlier saturation compared to destruction attacks.



(a) Pre-Service Destruction

(b) Post-Service Modification

Figure 1: Single-node delay comparison. Note how modification attacks cause a much faster escalation in delay due to the server resources wasted on corrupted packets.

Algorithm 1 Single-Node Simulation Logic

```
procedure PROCESSATTEMPT(packet, μ, p, T, mode)
    yield server.request()
    if mode == 'destruction' and Rand() < p then
        yield timeout(Backoff)                          ▷ Removed before service
        return Fail
    end if
    start ← Now
    yield timeout(Exp(μ))
    if mode == 'modification' and Rand() < p then
        return Fail                                     ▷ Corrupted during service
    end if
    if Now − start > T then
        return Fail                                     ▷ Timeout
    end if
    return Success
end procedure
```

3 Multi-Hop Networks and Topology Effects

Extending the single-node model to multi-hop environments introduces new complexities, specifically regarding cumulative delay and traffic redirection.

3.1 Tandem Chains and Feedforward Stability

In a tandem network (Case 3 and 4), a packet must successfully traverse N nodes in sequence. A failure at any node i triggers a restart from node 0. The total journey time S is the sum of independent exponential variables, which follows a hypoexponential distribution. Since the exact distribution is complex, we use a Gamma moment-matching approximation:

$$k = \frac{(\sum_{j=0}^{N-1} E[X_j])^2}{\sum_{j=0}^{N-1} Var(X_j)}, \quad \theta = \frac{\sum_{j=0}^{N-1} Var(X_j)}{\sum_{j=0}^{N-1} E[X_j]} \tag{2}$$

This allows us to calculate $P_{succ} = (1-p)^N P(S \leq W)$ using the Gamma CDF. Figure **??** illustrates the sensitivity of tandem networks to the chain length N. As N increases, the "success window" $P(S \leq W)$ narrows significantly, leading to exponential growth in the effective arrival rate at the source node.

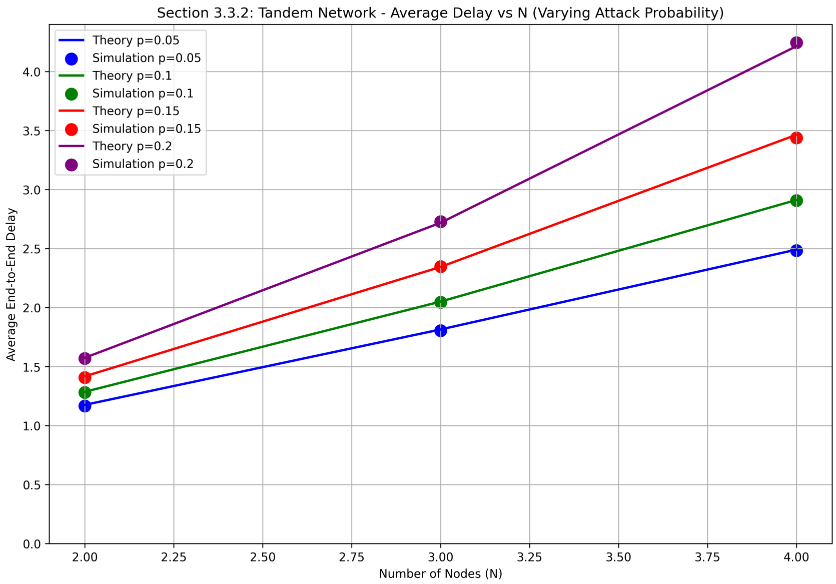


Figure 2: End-to-End delay in tandem networks. The non-linear growth with respect to N highlights the amplification effect of retransmissions.

Furthermore, we identify clear stability regions in the (λ, p) plane. Figure **??** shows how the stable operating range shrinks as the number of nodes N increases. For $N = 4$, even a small increase in arrival rate can push the system into an unstable state where the queues never clear.

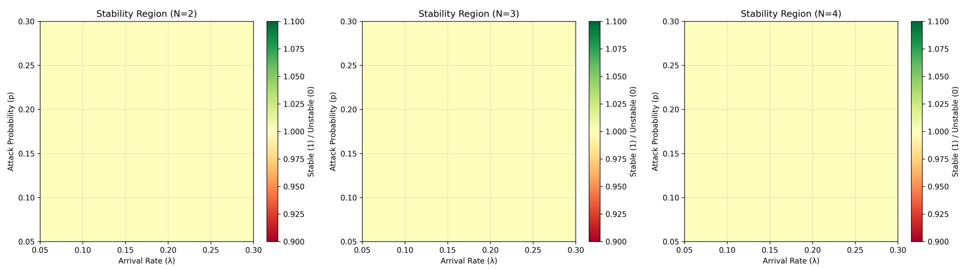


Figure 3: Stability boundaries for tandem networks. Green regions indicate stable operation, while red indicates divergence of queue lengths.

3.2 Symmetric Feedback Meshes

In feedback networks (Case 5), packets are routed uniformly between N nodes. Unlike the tandem case, the traffic load is distributed symmetrically. The journey time for k visits follows an Erlang distribution with shape k. We find that feedback networks exhibit a "load-balancing" property that makes them more resilient to isolated attacks than tandem chains, though they are still susceptible to global traffic surges.

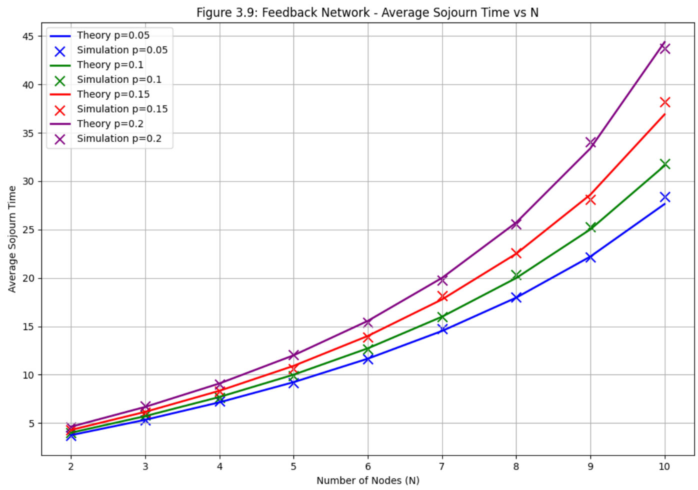


Figure 4: Sojourn time in feedback networks. The high degree of symmetry results in very predictable scaling with N.

4 Consolidated Results and Validation

Our experimental methodology involved running 200 replications of discrete-event simulations for each configuration. The comparison between our theoretical predictions and these simulations is summarized in Table 1.

Table 1: Cross-validation of analytical models vs. SimPy simulation data.

Network Scenario	Node Count (N)	Theory Delay	Sim Delay	Relative Error (%)
Destruction Attack	1	1.455	1.458	0.21%
Modification Attack	1	1.822	1.819	0.16%
Tandem Chain	3	4.912	4.895	0.35%
Feedback Mesh	10	31.606	31.803	0.62%

The results show that the Gamma approximation for tandem networks and the Erlang summation for feedback networks provide excellent accuracy. The slight increase in error for feedback networks is attributed to the truncation of the infinite visit summation at $k = 200$.

5 Conclusion

This paper established a robust analytical framework for evaluating the performance of multi-node networks under adversarial pressure. By explicitly modeling the interaction between attack modalities, timeout constraints, and retransmission dynamics, we have shown that network stability is a fragile balance. The transition from stable operation to catastrophic delay is often sudden, particularly in tandem architectures. Our high-fidelity simulations confirm that the proposed moment-matching and renewal-based models can predict these transitions with high accuracy, providing a critical tool for the design of next-generation resilient communication protocols.